Reg. No. | | | | | | | | | | |  [14CS/IS35]

# THIRD SEMESTER B.TECH EXAMINATION DECEMBER – 2015

## DATA STRUCTURES USING C

**Time:3 Hours** **Maximum Marks:70**

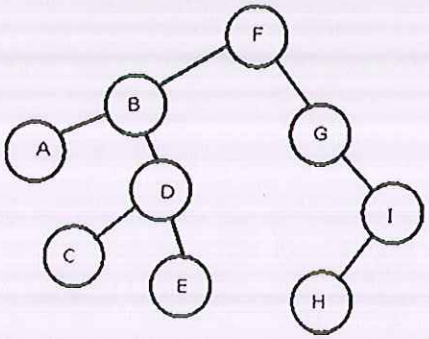
**Instruction:** Answer any **FIVE** full Questions, choosing at least **TWO** from each part.

### PART-A

| | | | |
|---|---|---|---|
| **1.** | **a.** | What is a pointer? Explain with an example. | **05** |
| | **b.** | Explain with example how to pass an array to a function. | **05** |
| | **c.** | Explain the following dynamic memory allocation function. | **04** |
| | | **i.** Malloc | |
| | | **ii.** Calloc | |
| **2.** | **a.** | What are strings? With example explain how to declare, read and print a string in C. | **07** |
| | **b.** | What is a structure? With example program explain the structure declaration, reading and printing. | **07** |
| **3.** | **a.** | Explain how to design recursive algorithm with an example. | **08** |
| | **b.** | Write a C program to print the Fibonacci series using recursion. | **06** |
| **4.** | **a.** | Define stack. Write a C program to implement a stack data structure. | **08** |
| | **b.** | List out the applications of stacks and queues. | **06** |

### PART-B

| | | | |
|---|---|---|---|
| **5.** | **a.** | Explain the ADT structure of lined list with a neat diagram. | **04** |
| | **b.** | Write a C program to insert, delete and traverse double linked list. | **10** |
| **6.** | **a.** | Perform the in-order, pre-order and post-order traversal for the given tree. | **06** |

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**b.** What are binary trees? Explain with an example. Also explain how to find the following properties of a binary tree. **08**

- **i.** Maximum height
- **ii.** Minimum height
- **iii.** Minimum nodes
- **iv.** Maximum nodes.

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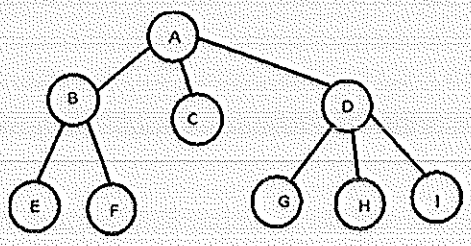
**7.** **a.** Define Heap. Explain min-heap, max-heap and priority queue. **06**

**b.** Explain how Heap is implemented using arrays with an example. **08**

**8.** **a.** What is a graph? What is the difference between directed and undirected graph? **04**

**b.** Explain the following with respect to graph with example: **06**

- **i.** Path
- **ii.** Cycle
- **iii.** Loop
- **iv.** Degree of a vertex

**c.** Perform the depth first and breadth first traversal for the following tree: **04**